Anmol Patlay

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Education

Maharashtra Institute of Technology, WPU **July. 2021 – July 2025**
Bachelor of Technology in ECE with specialization in AIML *Pune, Maharashtra*

Relevant Coursework

- Artificial Intelligence
- Machine Learning
- Deep Learning
- Optimization Techniques
- Python Programming
- Data Science
- Data Structures and Algorithms
- Computer Vision and Image Processing

Experience

Visteon Corporation **August 2025 – current**
Software Engineer *Pune, Maharashtra*

- Build a Python based backend automation framework that reduces manual testing effort and time by 70%.
- Collaborated with software and system teams to analyze complex issues, perform root cause analysis and deliver technical solutions.
- Developed reusable Python modules for hardware-software integration and automated validation workflows.
- Built a pipeline using FastAPI for getting live Execution data using Python listeners for getting real time data.

Projects

Job Application Tracker | *Python, FastAPI, Pydantic, JWT Authentication* **July-August 2026**

- Built a RESTful API(20 endpoints) with FastApi,SQLAlchemy ORM,and MySQL,using Pydantic schema for request/response validation for abstracting database models from client-facing interfaces and facilitating future schema evolution.
- Implemented JWT Authentication(OAuth2 password flow),bcrypt password hashing,dependency-injected auth guards,and user-scoped data access.
- Built secure account flows: registration,login,protected profile endpoint,and password change with old-password verification
- Introduced Rate-Limiters for every endpoint to prevent any brute-force attacks

California House Price Predictor | *FastAPI, Linear-Regression ,Docker, Render* **August 2026**

- Built and trained a regression model for house price prediction,serialized with joblib for fast and reusable inference.
- Developed a FastAPI backend exposing the model via REST endpoint,accepting input features and returning real-time price predictions.
- Containerized the application using Docker with a slim Python base image,structuring the build to leverage layer caching(dependencies installed before code copy) for faster rebuilds.
- Deployed the containerized service on Render via Github-integrated CI,enabling live public access to production API

AI in Agriculture-Image Segmentation for grapes | *Python , Tensorflow, Scikit-Learn, Deep-Learning* **2023**

- Developed a U-Net–based semantic segmentation model for grape cluster detection using the GrapesNet dataset in collaboration with IIT Hyderabad.
- Trained the model using T4-GPU, TensorFlow/Keras, leveraging Adam optimizer, Binary Cross-Entropy loss, data augmentation, and hyperparameter tuning.
- Achieved 97.93% Pixel Accuracy, 94.87% Dice Score, 90.25% IoU, and 0.0489 validation loss, demonstrating robust segmentation performance.
- Utilized Python, TensorFlow, Keras, OpenCV, NumPy, CUDA, Deep Learning, Semantic Segmentation, Computer Vision, and Image Processing to deliver a high-performance research solution.

Technical Skills

Languages: Python, C++, SQL
AI/ML: scikit-learn, TensorFlow, PyTorch, U-Net, CNN, Lang Chain, HuggingFace Transformers, OpenAI API, RAG pipelines, Prompt Engineering
Backend/APIs: FastAPI, Flask, REST APIs, SQLAlchemy, MySQL, Redis (basics)
SQL/Databases:MySQL, Relational Database Design and Internals, Complex Joins, CTEs, Window Functions, Views, Normalization, Query Optimization, Indexing, Transactions (ACID),System design
Devops/Tools: Docker, Git, GitHub Actions (CI/CD), Linux CLI